

FUTENMA MCAS, Japan

WMO: 479330

Lat: 26.274N

Lon: 127.756E

Elev: 78

StdP: 100.39

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 42215

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.0	11.9	2.8	4.7	12.9	3.8	5.0	13.3	11.3	13.8	10.4	14.7	4.5	10	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	32.4	26.6	32.0	26.6	31.2	26.4	28.1	30.3	27.6	29.9	27.2	29.6	5.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.4	29.4	26.9	22.8	29.1	26.3	22.0	28.8	91.0	30.6	88.5	30.1	86.9	29.8	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.6	8.5	9.3	34.0	1.3	1.3	8.4	34.9	7.6	35.7	6.9	36.4	6.0	37.3		
			6.2	29.3	1.1	0.8	5.4	29.9	4.7	30.4	4.1	30.9	3.3	31.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.0	16.8	16.9	18.5	21.2	24.0	27.1	29.1	29.0	27.7	25.1	22.0	18.4
	DBStd	4.93	2.44	2.68	2.78	2.62	2.08	2.04	1.09	1.07	1.42	1.88	2.10	2.39
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	187	61	56	33	6	0	0	0	0	0	0	1	29
	CDD10.0	4745	211	193	263	336	433	514	592	588	530	467	358	261
	CDD18.3	1890	13	16	38	91	174	264	333	329	280	209	109	32
	CDH23.3	16702	4	6	44	266	957	2639	4089	3971	2948	1435	320	23
	CDH26.7	5257	0	0	1	32	133	781	1656	1543	870	229	13	0
Wind	WSAvg	4.4	4.2	4.4	4.5	4.5	4.3	4.9	4.6	4.5	4.4	4.4	4.3	4.3
Precipitation	PrecAvg	2078	121	110	153	165	244	256	191	248	181	159	108	139
	PrecMax	3135	309	406	367	460	691	570	620	651	459	585	260	822
	PrecMin	1034	27	28	20	21	16	22	12	38	44	4	4	23
	PrecStd	544	56	83	90	99	140	139	171	148	97	123	75	140
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	24.1	26.1	29.2	30.5	32.6	33.5	33.3	32.3	30.9	28.1	25.2
		MCWB	20.1	20.6	21.8	24.3	25.7	27.0	26.8	26.8	26.4	25.6	24.0	21.4
	2%	DB	22.8	23.0	24.5	27.2	29.1	31.8	32.5	32.2	31.2	29.4	26.9	24.0
		MCWB	19.3	19.9	21.1	23.6	25.1	26.7	26.5	26.7	26.1	24.9	23.0	20.3
	5%	DB	21.4	22.0	23.3	26.0	28.1	31.0	32.0	31.8	30.9	28.8	26.0	23.1
		MCWB	17.9	19.0	20.3	22.6	24.6	26.4	26.5	26.6	26.0	24.5	22.1	19.4
	10%	DB	20.8	21.1	22.2	24.8	27.1	30.1	31.2	31.1	30.0	28.0	25.1	22.1
		MCWB	17.3	18.4	19.5	21.9	24.0	26.1	26.4	26.5	25.7	24.0	21.6	18.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	22.1	22.9	25.3	26.8	28.6	29.2	28.6	28.0	27.1	25.0	22.7
		MCDB	22.8	23.3	24.5	28.4	28.5	30.7	31.4	31.0	30.1	28.6	26.7	24.0
	2%	WB	20.0	20.9	21.9	24.2	26.0	27.8	28.1	27.8	27.3	26.2	24.0	21.5
		MCDB	22.0	22.2	23.6	26.4	28.1	29.8	30.5	30.3	29.5	28.0	25.8	23.0
	5%	WB	18.8	19.8	21.2	23.4	25.4	27.2	27.5	27.3	26.9	25.6	23.2	20.4
		MCDB	20.7	21.3	22.8	25.3	27.4	29.4	30.1	29.9	29.2	27.6	25.0	22.2
	10%	WB	17.8	18.8	20.2	22.6	24.7	26.7	27.1	27.0	26.5	25.0	22.6	19.4
		MCDB	20.0	20.6	22.1	24.3	26.7	28.9	29.7	29.7	28.8	27.2	24.5	21.6
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.0	5.3	5.2	4.8	4.5	4.7	4.5	4.7	4.9	5.0	5.0
		MCDBR	5.7	5.3	5.7	5.7	5.4	5.0	5.2	5.2	5.2	5.3	5.2	5.5
		MCWBR	4.3	4.3	4.3	3.3	3.2	2.5	2.5	2.4	2.6	2.9	3.4	3.8
	5% WB	MCDBR	4.9	4.8	5.4	5.1	5.0	4.7	4.8	4.6	4.5	4.5	4.6	4.7
		MCWBR	4.6	4.6	4.8	3.6	3.7	2.9	2.9	2.8	2.8	3.0	3.5	4.2
Clear-Sky Solar Irradiance	taub	0.455	0.484	0.557	0.565	0.509	0.469	0.469	0.487	0.482	0.480	0.463	0.454	
	taud	2.138	2.076	1.905	1.946	2.169	2.350	2.343	2.281	2.264	2.209	2.213	2.147	
	Ebn at Noon	779	788	753	759	800	828	827	813	808	784	768	759	
	Edh at Noon	136	155	193	190	152	126	127	134	134	134	125	130	
All-Sky Solar Radiation	RadAvg	2.43	3.07	3.79	4.42	4.93	5.45	6.51	5.96	5.22	4.10	3.00	2.45	
	RadStd	0.29	0.46	0.36	0.44	0.64	0.49	0.53	0.40	0.39	0.36	0.28	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	+0.41	+0.53	N/A	N/A	N/A	N/A

Nomenclature: See separate page